

A Social Choice Analysis of Retroactive Funding

GovXS (Eyal Briman, Angela Kreitenweis, Muhammad Idrees, Nimrod Talmon)

February 12, 2025

Abstract

Optimism Retro Funding offers a sustainable model for individuals to receive rewards for their contributions to Optimism’s success, ensuring that efforts are not only recognized but also effectively rewarded.

Retroactive Funding is based on the idea that it’s easier to agree on what was useful in the past than what might be useful in the future.

This project aims to measure how different Retro Funding voting designs perform against a number of requirements aimed at optimizing different objectives. We conduct a social choice analysis of the mechanisms used for OP Retro Funding Round 1–4 and provide a comparison of the voting rules in place. Finally we suggest design improvements to optimize towards the objectives defined.

1 Introduction

We provide a social choice analysis of the social choice mechanisms used for Optimism Retro Funding. The philosophy is to have an economic model for public goods. Historically, Optimism is doing now the 4th installation.¹

1.1 Informal Goals

We are interested in comparing various voting rules for retrofunding. Informally, our goals are to compare the rules to better understand how the various rules behave with respect to:

- Resistance to malicious behavior: Measure how a malicious voter can impact voting outcomes
- Resistance to collusion: Measure how a group of coordinated malicious voters can impact voting outcomes
- Incentive compatibility: Prove or disprove that the voting design is incentive compatible (“every participant can achieve their own best outcome by acting according to their true preferences”)
- Simplicity for voters: Voters can easily understand how the voting design works and how to best behave to achieve their goals
- Expected Outcome: the voting rule does not create unexpected outcomes
- Result maximizes Social Welfare (or represents the preference of the majority of voters)
- Result represents diversity (results are appropriate wrt. voter preferences)
- Incentives of voters and OP Collective are aligned
- Impact = Profit (Fair Share)

2 Related Work

We discuss the retrofunding setting from a social choice perspective; then we consider objective social choice (and epistemic democracy) as they relate to the setting of retrofunding.

¹Retro 1: <https://medium.com/ethereum-optimism/retropgf-experiment> Retro 2: <https://optimism.mirror.xyz/7v1DehEY3dpRcYFhqWrVnc9Qj94H2L976LK1WH1FX-8> Retro 3: <https://gov.optimism.io/t/retropgf-3-round-design/6802> Retro 4: <https://optimism.mirror.xyz/7v1DehEY3dpRcYFhqWrVnc9Qj94H2L976LK1WH1FX-8>

2.1 Retrofunding Setting

From voting theory perspective, it is very similar to portioning.² An interesting bit is metric-based voting.^{3 4}

2.2 Objective Social Choice and Epistemic Democracy

Note also that it is somewhat objective in nature.

3 Formal Model

We provide a formal model for (token-based) retrofunding.

3.1 Definitions

- $N = \{1, 2, \dots, n\}$: Set of voters.
- $P = \{1, 2, \dots, m\}$: Set of projects.
- $B \in \mathbb{N}$: Total budget available for distribution, normalized to 1.
- $c \in \mathbb{N}$: The amount of tokens given to each voter to distribute among the projects (under the constraint of $c \cdot n \leq B$), normalized to 1.

3.2 Budget Allocations

An allocation $\mathbf{x} = \{a_1, \dots, a_m\}$ represents the distribution of tokens among projects. Each project p receives a_p tokens, where $a_p \in [0, 1]$. We denote A as a the group of all feasible allocations $\mathbf{x} = \{a_1, \dots, a_m\}$, such that $\sum_{p \in P} a_p \leq 1$.

3.3 Ballot Design and Vote Aggregation

Each voter is given k tokens to distribute among the projects, a mechanism referred to in the literature as *cumulative ballot*. Formally we denote $x_{i,p} \in \mathbb{R}$

to represent the amount of tokens voter i allocates to project p , subject to the constraint:

$$\sum_{p \in P} x_{i,p} \leq 1$$

and $X_i \in \mathbb{R}^m$ is the vector of funds distribution provided by voter i . The aggregation of votes (i.e., funds distribution) is given by a voting rule now presented formally as a function:

$$f : (\{X_1, \dots, X_n\}) \rightarrow \{a_1, \dots, a_m\} = \mathbf{x}$$

ensuring that the total budget constraint is respected:

$$\sum_{p \in P} a_p \leq 1,$$

Remark 1. Notice that we model the case where the entire budget is allocated:

$$\sum_{p \in P} a_p = 1,$$

but voters are not required to allocate all their c tokens:

$$\sum_{p \in P} x_{i,p} \leq 1.$$

²<https://arxiv.org/pdf/2003.00606>

³<https://gov.optimism.io/t/experimentation-impact-metric-based-voting/7727>

⁴<https://vitalik.eth.limo/general/2021/11/16/retro1.html>

However, there is a strategic advantage for voters to allocate all their tokens, as this maximizes their voting power. Therefore, we can assume:

$$\sum_{p \in P} x_{i,p} = 1.$$

4 Preliminaries

In this section, we provide some needed preliminaries. In particular, we speak of objective social choice (compared to subjective social choice); as well as discuss various, useful distance measures.

4.1 Objective Social Choice

In objective social choice, the objective truth refers to a "correct" or "ideal" outcome $\mathbf{x}^* \in A$ that is independent of voters' preferences. This outcome, often referred to as the ground truth, is predetermined and not necessarily the result of maximizing a utility function.

Formal Definition Let A be the set of all feasible allocations, where each allocation $\mathbf{x} \in A$ is a vector representing the distribution of resources among m projects such that:

$$\mathbf{x} = \{a_1, a_2, \dots, a_m\}, \quad a_i \geq 0 \quad \text{and} \quad \sum_{i=1}^m a_i = B$$

where B is the total budget.

The objective truth $\mathbf{x}^*$ is a fixed, externally defined allocation that represents the optimal outcome for the community, based on criteria outside the voters' preferences.

4.1.1 Voters as Noisy Estimates of the Truth

Voters are considered as providing noisy estimates of the objective truth $\mathbf{x}^*$. Each voter v submits an allocation $\mathbf{x}_v$, which is assumed to be close to $\mathbf{x}^*$ within some error margin.

A useful formalism is to define the distance between a voter's allocation and the objective truth using a distance metric d (e.g., ℓ_1 norm):

$$d(\mathbf{x}_v, \mathbf{x}^*) = \|\mathbf{x}_v - \mathbf{x}^*\|_1$$

The goal is to find a voting rule f that aggregates the noisy ballots $\{\mathbf{x}_v\}_{v=1}^n$ to produce an allocation $\mathbf{x}_{\text{agg}}$ that minimizes the distance to the objective truth:

$$\mathbf{x}_{\text{agg}} = f(\{\mathbf{x}_v\}_{v=1}^n)$$

such that

$$d(\mathbf{x}_{\text{agg}}, \mathbf{x}^*) \text{ is minimized.}$$

4.2 Distance Metrics

Consider two vectors $\mathbf{a} = (a_1, a_2, \dots, a_m)$ and $\mathbf{b} = (b_1, b_2, \dots, b_m)$. We define the following distance metrics:

4.2.1 ℓ_1 Distance

The ℓ_1 distance (also known as Manhattan distance or taxicab distance) between the vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as:

$$\ell_1(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|_1 = \sum_{i=1}^m |a_i - b_i|.$$

5 Voting Rules

We consider the following voting rules:

- **OP RetroPGF Round 1-Quadratic Voting:**

Quadratic Voting (QV) allows voters to express the intensity of their preferences by purchasing votes where the cost of votes increases quadratically. In the case of OP RetroFunding, voters receive a predefined amount of tokens for voting. Hence, in Quadratic Voting for RetroFunding, we calculate the vote distribution so that the properties of the Quadratic Voting rule (quadratic cost) are met.

- Each voter $i \in N$ votes and distributes $x_{i,p}$ tokens to project p .
- The true vote, the tokens distributed to each $x_{i,p}$ tokens is actually $\sqrt{x_{i,p}}$.
- The vote distribution to project p is the sum of the tokens distributed by all voters:

$$a_p = \frac{\sum_{i \in N} \sqrt{x_{i,p}}}{\sum_{i \in N} \sum_{p' \in P} \sqrt{x_{i,p'}}}$$

- **OP RetroPGF Round 2- R2 Mean Rule:** In Weighted Average voting (Mean rule), each voter's allocation to a project is weighted by their total allocation.

- Each voter $i \in N$ allocates $x_{i,p}$ tokens to project p .
- The weight of project p is calculated as:

$$a_p = \frac{\sum_{i \in N} x_{i,p}}{\sum_{i \in N} \sum_{p' \in P} x_{i,p'}}$$

- **OP RetroPGF Round 3- R3 Quorum Median:** R3 Quorum Median ensures that the allocation is based on the median of the voters' allocations, adjusted by a scale factor.

- Each voter $i \in N$ allocates $x_{i,p}$ tokens to project p .
- A quorum is defined as the minimal amount of tokens needed q_1 for a project and the minimal amount of votes q_2 needed for a project to get funded.
- For each project p , the median allocation a_p is calculated as:

$$b_p = \frac{\text{median}\{x_{i,p} | x_{i,p} > 0, i \in N\}}{\sum_{p' \in P} \text{median}\{x_{i,p'} | x_{i,p'} > 0, i \in N\}}$$

$$d_p = \begin{cases} b_p & \text{if } b_p \geq q_1 \text{ AND } \sum_{i \in N} c_{i,p} > q_2 \\ 0 & \text{else} \end{cases}$$

Having

$$c_{i,p} = \begin{cases} 1, & \text{if } x_{i,p} > 0 \\ 0, & \text{else} \end{cases}.$$

Finally:

$$a_p = \frac{d_p}{\sum_{p' \in P} d_{p'}}$$

- **OP RetroFunding Round 4- R4 Median Impact Metric Score:**

Each voter $i \in N$ gives a set of m weights denoted by $\{w_{i,1}, \dots, w_{i,m}\}$ expressing the importance of each metric.

- * The impact metric share of metric $j \in [m]$ for each project $p \in P$ is calculated as:

$$S_{j,p} = \left(\frac{s_{j,p} \cdot m_p}{\sum_{p' \in P} s_{j,p'} \cdot m_{p'}} \right)$$

where $S_{j,p}$ is the impact metric value provided by the project in the application and m_p is the multiplier for project p . If p is an open source project, $m_p > 1$ equals the open source multiplier; otherwise, m_p equals 1.

* The total score for each project p by a voter i is calculated as:

$$\text{Score}(i, p) = \sum_{j \in M} (S_{j,p} \cdot w_{i,j})$$

Next, the process continues to,

Capping: The score for each project p is capped at a maximum of K_1 tokens. If

$$\text{Score}(i, p) > K_1,$$

the excess amount is redistributed proportionally among the remaining projects of the same voter.

The capped score for each project p given a constant of K_1 tokens is defined as:

$$\text{Capped Score}(i, p) = \min(\text{Score}(i, p), K_1).$$

Redistribution: Any excess tokens from the capping is redistributed to the other projects of the same voter proportionally to their uncapped scores. Let the total excess amount for voter i be denoted by

$$\text{Excess}(i) = \sum_{p \in P} \max(\text{Score}(i, p) - K_1, 0).$$

The redistributed score for each project p is then:

$$x_{i,p} = \text{Capped Score}(i, p) + \frac{\text{Excess}(i) \cdot \text{Score}(i, p)}{\sum_{p' \in P'} \text{Score}(i, p')}.$$

Having $P' = \{p \in P | \text{Score}(i, p) < K_1\}$

* Finally, the funding allocation is calculated as:

$$c_p = \frac{\text{median}\{x_{1,p}, x_{2,p}, \dots, x_{n,p}\}}{\sum_{p' \in P} \text{median}\{x_{1,p'}, x_{2,p'}, \dots, x_{n,p'}\}} \cdot B$$

We then cap the median score for each project with maximum of K_2 tokens and redistribute remaining tokens:

$$d_p = \min(c_p, K_2) + \frac{\sum_{p \in P} \max(0, c_p - K_2) \cdot c_p}{\sum_{p' \in P'} c_p}.$$

Having $P' = \{p \in P | c_p < K_2\}$. Then we normalise: $d_p := \frac{d_p}{\sum_{p \in P} d_p} \cdot B$.

Then we continue to omit projects that are allocated with less than K_3 tokens and redistribute the removed projects token allocation:

$$b_p = \begin{cases} 0, & \text{if } d_p < K_3 \\ d_p + \frac{\sum_{p' \in \{d_{p''} < K_3 | p'' \in P\}} d_{p'}}{\sum_{p' \in \{d_{p''} \geq K_3 | p'' \in P\}} d_{p'}} \cdot d_p, & \text{else} \end{cases}$$

Finally we normalize one last time to get the final allocation:

$$a_p = \frac{b_p}{\sum_{p' \in P} b_{p'}}.$$

• OP RetroFunding Round 4- R4a Simplified Capped Median :

In the simplified version of the Round4 voting rule, we skip processing impact metric values to impact metric scores. Hence, we start a simplified voting input:

Each voter $i \in N$ allocates $x_{i,p} \in [K1]$ tokens to project p .

The funding allocation is calculated as:

$$c_p = \frac{\text{median}\{x_{1,p}, x_{2,p}, \dots, x_{n,p}\}}{\sum_{p' \in P} \text{median}\{x_{1,p'}, x_{2,p'}, \dots, x_{n,p'}\}} \cdot B$$

We then cap the median score for each project with maximum of K_2 tokens and redistribute remaining tokens:

$$d_p = \min(c_p, K_2) + \frac{(\sum_{j \in P} \max(0, c_j - K_2)) \cdot c_p}{\sum_{j \in P} c_j}.$$

Then we normalise: $d_p := \frac{d_p}{\sum_{p \in P} d_p} \cdot B$.

Then we continue to omit projects that are allocated with less than K_3 tokens and redistribute the removed projects token allocation:

$$b_p = \begin{cases} 0, & \text{if } d_p < K_3 \\ d_p + \frac{(\sum_{p' \in \{d_{p''} < K_3 | p'' \in P\}} d_{p'})}{\sum_{p' \in \{d_{p''} \geq K_3 | p'' \in P\}} d_{p'}} \cdot d_p, & \text{else} \end{cases}$$

Finally we normalize one last time to get the final allocation:

$$a_p = \frac{b_p}{\sum_{p' \in P} b_{p'}}.$$

Adapted Voting Rules-design proposals to avert vulnerabilities/weaknesses we find in the evaluation. We consider the following voting rules:

- **Normalized Median Rule**-The normalized median rule computes the median allocation for each project:

$$a_p = \frac{\text{median}\{x_{1,p}, x_{2,p}, \dots, x_{n,p}\}}{\sum_{p' \in P} \text{median}\{x_{1,p'}, x_{2,p'}, \dots, x_{n,p'}\}}$$

for each project $p \in P$.

- **Midpoint Rule** [?]- The Midpoint Rule selects the input allocation (i.e., the ballot of one of the voters) that minimizes the ℓ_1 distance from all other ballots.

$$\text{MR}(X_i) = \sum_{j=1}^n \|X_i - X_j\|_1$$

The chosen allocation is:

$$\mathbf{x} = X_{i^*} \quad \text{where} \quad i^* = \arg \min_{i \in N} \text{MR}(X_i)$$

In this context, the ℓ_1 distance between two allocations X_i and X_j is given by:

$$\|X_i - X_j\|_1 = \sum_{p \in P} |x_{i,p} - x_{j,p}|$$

- **Moving Phantoms**- A phantom system $\mathcal{F}$ is a family of functions $\{f_k : k \in \{0, \dots, n\}\}$ where each $f_k : [0, 1] \rightarrow [0, 1]$ is continuous, weakly increasing, and satisfies the following conditions:

- $f_k(0) = 0$
- $f_k(1) = 1$
- $f_0(t) \geq f_1(t) \geq \dots \geq f_n(t)$ for all $t \in [0, 1]$

The moving phantom mechanism $\mathcal{A}^F$ is defined for all voters and projects as:

$$\mathcal{A}^F(a_p) = \text{med}(f_0(t^*), \dots, f_n(t^*), x_{1,p}, \dots, x_{n,p})$$

where t^* is chosen such that ⁵:

$$t^* \in \left\{ t : \sum_{p \in P} \text{med}(f_0(t), f_1(t), \dots, f_n(t), x_{1,p}, \dots, x_{n,p}) = 1 \right\}.$$

We consider two types of phantom based mechanisms:

⁵We can easily find t^* using a simple binary search.

1. **The Independent Markets Algorithm-** The Independent Market algorithm denoted by $\mathcal{A}^{IM}$ is a moving phantom algorithm given by:

$$f_k(t) = \min\{t(n - k), 1\}$$

2. **Majoritarian Phantoms Algorithm-** this algorithm denoted by $\mathcal{A}^{F*}$ is given by:

$$f_k(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{k}{n+1}, \\ t(n+1) - k & \text{if } \frac{k}{n+1} < t \leq \frac{k+1}{n+1}, \\ 1 & \text{if } \frac{k+1}{n+1} \leq t \leq 1. \end{cases}$$

6 Evaluation Metrics

We define the evaluation metrics :

- **Resistance to malicious behavior:**

1. **Bribery-** Consider a vote profile $V = \{X_1, \dots, X_n\}$ and its outcome $\mathbf{x} = \{a_1, \dots, a_m\}$ with respect to some voting rule f . Let c be the total amount of tokens each voter can distribute among the projects. The cost to move a token from one project to another is 1.

Now, let p be a special project, and X be the desired increase in funds allocated to project p . The goal is to determine the minimum budget b needed for a bribery strategy that results in a new vote profile $V' = \{X'_1, \dots, X'_n\}$ such that the new outcome $\mathbf{x}' = \{a'_1, \dots, a'_m\}$ with respect to the same voting rule f assigns X more tokens to project p compared to the original outcome $\mathbf{x}$.

Formally, we define the bribery cost metric as follows:

Given the original outcome:

$$\mathbf{x} = f(V) = \{a_1, \dots, a_m\}$$

Determine the minimum b such that there exists a bribed profile $V' = \{X'_1, \dots, X'_n\}$ resulting in:

$$\mathbf{x}' = f(V') = \{a'_1, \dots, a'_m\}$$

and

$$a'_p = a_p + X$$

The total cost b is computed as:

$$b = \sum_{i=1}^n \sum_{q \in P} |x_{i,q} - x'_{i,q}|$$

where $x_{i,q}$ and $x'_{i,q}$ are the amounts of tokens voter i allocates to project q in profiles V and V' , respectively.

This metric measures the minimum effort required to manipulate the vote profile such that the funds allocated to the special project p increase by X . We are interested in:

- The theoretical minimum bribery cost.
 - The theoretical maximum bribery cost.
 - The average bribery cost with respect to some distribution.
 - The expected needed bribery cost for a given dataset.
2. **Cost of Control-** Consider a vote profile $V = \{X_1, \dots, X_n\}$ and its outcome $\mathbf{x} = \{a_1, \dots, a_m\}$ with respect to some voting rule f . Let c be the total amount of tokens each voter can distribute among the projects.

Now, let p be a special project, and r be the desired percentage increase in funds allocated to project p . The goal is to determine the minimum number of voters needed to change the vote profile V by adding or removing voters (we assume removing or adding a voter is equivalent to 1 currency), such that the new outcome $\mathbf{x}'$ changes in favor of the manipulator, meaning the fund of project p increases by r percent.

We acknowledge the work of Bartholdis et al. (1992) [?] that presents resistance to control of voting rules (whether by adding or removing projects/votes) by examining the computational complexity of each pair of voting rules and types of control.

- **Incentive compatibility:**

1. **Strategyproofness-** We assume strategyproofness for a mechanism when every participant can achieve their own best outcome by acting according to their true preferences. With a strategyproof voting rule, voters would not manipulate their votes to improve their own outcome.

Formally, consider a voter $i \in N$. Fix the votes of all other voters. Let $u(X_i)$ represent the utility of voter i when voting truthfully, and $u(X'_i)$ represent the utility of voter i when voting differently. For the mechanism to be strategyproof, the following condition must hold:

$$u(X_i) \geq u(X'_i)$$

This means that the utility of voter i when voting truthfully X_i is always greater than or equal to the utility when voting strategically X'_i .

2. **Group-strategyproofness-** A mechanism is group-strategyproof if no group of agents can misvote such that all its members are strictly better off. This extends the concept of strategyproofness to consider the possibility of a group of voters colluding to improve their outcomes. It is important to note that not all members of the colluding coalition need to gain directly, as they can make in-coalition payments after collusion to settle the gains.

Formally, consider a coalition of voters $C = \{X_1, X_2, \dots, X_k\}$. Let X_{-C} be the votes of all voters not in the coalition. Let $u(X_i)$ represent the utility of voter X_i when voting truthfully, and $u(X'_i)$ represent the utility of voter X_i when the coalition votes strategically. For the mechanism to be group-strategyproof, the following condition must hold for all coalitions C and all $X_i \in C$:

$$\exists X_i \in C \text{ such that } u(X_i) > u'(X_i)$$

This means that there is at least one member in any coalition C whose utility is not strictly better off when the coalition votes strategically compared to when they vote truthfully.

- **Participation-** This criterion ensures that every voter has an incentive to participate in the voting process. Formally, we consider a vote X_i and let $\mathbf{x} = \{a_1, \dots, a_m\}$ be the outcome of the election without vote X_i , and $\mathbf{x}' = \{a'_1, \dots, a'_m\}$ be the outcome with vote X_i . Participation criterion is satisfied if for all voters $i \in N$ and for all possible outcomes $\mathbf{x} \in A$, the following condition holds:

$$u(\mathbf{x}') \geq u(\mathbf{x}).$$

- **Expected Outcome-** We want the voting designs to behave as expected and to be able to control abnormalities of voters. We use the following metrics:

1. **Robustness-** This metric evaluates the influence a single voter can have on the outcome and how robust the results of committee elections are to small changes in the input preference orders, depending on the voting rules used. The goal is to measure the distance in the output when a voter's vote is randomly changed, simulating a randomized version of bribery. It is also interesting to see the maximum change possible.

Formal Definition:

Consider a vote profile $V = \{X_1, \dots, X_n\}$ and its outcome $\mathbf{x} = \{a_1, \dots, a_m\}$ with respect to some voting rule f . Let X_i represent the vote of a single voter. Let X'_i represent a randomly changed vote of X_i . Let $\mathbf{x}' = f(V')$ be the new outcome where $V' = \{X_1, \dots, X_{i-1}, X'_i, X_{i+1}, \dots, X_n\}$.

We measure the distance between the original outcome $\mathbf{x}$ and the new outcome $\mathbf{x}'$ using a distance metric d . Common choices for d include the ℓ_1 and ℓ_2 distances. The robustness metric is defined as the expected value of the distance when a single vote is changed:

$$\text{Robustness} = \mathbb{E}[d(\mathbf{x}, \mathbf{x}')]]$$

where the expectation is taken over all possible changes to the vote X_i .

To empirically evaluate robustness, one can simulate the process as follows:

- (a) Take the original vote profile V and compute the outcome $\mathbf{x} = f(V)$.
 - (b) Randomly change the vote of a single voter X_i to obtain X'_i .
 - (c) Compute the new outcome $\mathbf{x}' = f(V')$.
 - (d) Calculate the distance $d(\mathbf{x}, \mathbf{x}')$.
 - (e) Repeat the process multiple times and compute the average distance.
2. **Voter Extractable Value (VEV)**- For each voter and project in a given instance and voting rule, we examine the final allocation sensitivity when a voter allocates 90-99% of their available funding, c , to a specific project and distributes the remainder equally across all other projects. We calculate the maximum ℓ_1 distance from the original allocation for every voter and project, representing the greatest skewness possible under that voting rule. Finally, we compute the average maximum skewness across many instances.

Formal Definition:

Consider a vote profile $V = \{X_1, \dots, X_n\}$ and its outcome $\mathbf{x} = \{a_1, \dots, a_m\}$ with respect to some voting rule f . Let X_i represent the vote of voter i , where each voter i allocates their available funding c across m projects. Let $X_i^{(k)}$ represent a modified version of X_i where the voter allocates $r\%$ of their funding c to project $k \in [m]$, and distributes the remaining $(1-r)\%$ equally across all other $m-1$ projects. Let $\mathbf{x}^{(i,k)} = f(V^{(i,k)})$ represent the new outcome, where $V^{(i,k)} = \{X_1, \dots, X_{i-1}, X_i^{(k)}, X_{i+1}, \dots, X_n\}$ is the vote profile after this change. The outcome with respect to voting rule f is now denoted by: $\mathbf{x}^{(i,k)} = \{a_1^{(i,k)}, \dots, a_m^{(i,k)}\}$. We measure the distance between the original outcome $\mathbf{x}$ and the new outcome $\mathbf{x}^{(i,k)}$ using the ℓ_1 -distance:

$$d(\mathbf{x}, \mathbf{x}^{(i,k)}) = \sum_{j=1}^m |a_j - a_j^{(i,k)}|$$

The **VEV** for an instance I (i.e., n votes to m projects) and a voting rule is set as:

$$\text{VEV}(I) = \max_{i \in [n], k \in [m]} d(\mathbf{x}, \mathbf{x}^{(i,k)})$$

3. **Pareto Efficiency**- Let $V = \{v_1, v_2, \dots, v_n\}$ be a profile of voters and $f(V) = p$ be the output of the rule f for this profile. We say that f is Pareto efficient if there does not exist an output x' such that:

$$\forall v_i \in V, U(v_i, x') \geq U(v_i, x)$$

and

$$\exists v_j \in V \text{ such that } U(v_j, x') > U(v_j, x)$$

where $U(v_i, x)$ denotes the utility of voter v_i for the outcome x .

In other words, a voting rule f satisfies Pareto efficient if it holds that, for each voting profile V with the winner p , there is no other solution x' such that all voters are as happy with x' and at least one voter is happier (i.e., has smaller ℓ_1 distance)

4. **Monotonicity:**

A voting rule satisfies monotonicity if, when a voter reallocates tokens in favor of a project, the allocation to that project does not decrease.

Formal Definition:

Consider a vote profile $V = \{v_1, \dots, v_n\}$, where v_i represents the preference of voter i . Let $f(V) = \mathbf{x} = \{a_1, \dots, a_m\}$ be the allocation of tokens to projects under the voting rule f , where a_p represents the allocation to project p .

Let V' be a new vote profile obtained from V by changing the vote of one voter j such that the voter j reallocates more tokens to project p by taking tokens away from other projects. Formally, this means:

$$\begin{aligned} v_j &= (x_{j,1}, \dots, x_{j,p}, \dots, x_{j,m}) \quad \text{in } V \\ v'_j &= (x'_{j,1}, \dots, x'_{j,p}, \dots, x'_{j,m}) \quad \text{in } V', \end{aligned}$$

where $x'_{j,p} > x_{j,p}$ and $x'_{j,p'} \leq x_{j,p'}$ for $p' \neq p$, ensuring that the additional tokens allocated to project p are taken from the allocations to other projects, and the total tokens distributed by voter j remain the same:

$$\sum_{p \in P} x'_{j,p} = \sum_{p \in P} x_{j,p}.$$

A voting rule f satisfies **monotonicity** if the following condition holds:

$$f(V, p) \leq f(V', p),$$

meaning the allocation to project p does not decrease when a voter reallocates more tokens to it by reducing their allocation to other projects.

5. Reinforcement:

Consider two vote profiles $V_1 = \{x_{1,1}, \dots, x_{1,m}\}$ and $V_2 = \{x_{2,1}, \dots, x_{2,n}\}$ where each $x_{i,j}$ represents the allocation of tokens by voter i to project j in profiles V_1 and V_2 , respectively. A voting rule f satisfies **reinforcement** if given $f(V_1) = f(V_2)$ then:

$$f(V_1 \cup V_2) = f(V_1) = f(V_2)$$

6. Proportionality:

In a cumulative voting system for budget allocation, proportionality ensures that if a group of voters, each distributing their votes entirely to a single project, makes up at least a fraction $\frac{1}{m}$ of the electorate, then this group can guarantee that at least $\frac{1}{m}$ of the total budget is allocated to their preferred project.

Formal Definition:

Consider a vote profile $V = \{v_1, \dots, v_n\}$, where each voter v_i allocates C tokens among P projects, and the total budget to be allocated is B . Let $S \subseteq V$ be a group of voters where each voter $v_i \in S$ allocates all C tokens to a single preferred project, making their vote profile uniformly focused on one project. Suppose this group S constitutes at least a fraction $\frac{1}{m}$ of the electorate, meaning $|S| \geq \lceil \frac{1}{m} \times n \rceil$.

A voting rule f satisfies **proportionality** if the following condition holds:

For each project j , the budget allocated to j is proportional to the number of voters who have allocated all their tokens to j . Specifically, if $\text{Appr}(j)$ denotes the set of voters who allocate all their tokens to project j :

$$f(j) = \frac{|\text{Appr}(j)|}{n} \times B,$$

where $f(j)$ represents the budget allocated to project j , $|\text{Appr}(j)|$ is the number of voters who have allocated all their tokens to j , and B is the total available budget.

This ensures that if a cohesive group of voters, who allocate their votes entirely to one project, forms a sufficient portion of the electorate, they can ensure a proportionate share of the budget is allocated to their preferred project.

- **Utilitarian Social Welfare-** The allocation maximizes Social Welfare and thus represents the preference of the majority of voters. The metric used here is the average ℓ_1 distance of the outcome $\mathbf{x}$ from all votes X_i formally:

$$\frac{1}{n} \cdot \sum_{i=1}^n \|\mathbf{x} - X_i\|_1$$

- **Diversity-** We use two metrics:

1. **Gini Index of Agents-** The Gini Index measures the inequality among values of a frequency distribution. In our setting it is applied to determine the amount of inequality among the utilities of voters denoted as the ℓ_1 distances from a voter ballot and the output.

$$\text{Gini Index of Agents} = \frac{\sum_{i=1}^n \sum_{j=1}^n \left| \|\mathbf{x}, X_i\|_1 - \|\mathbf{x}, X_j\|_1 \right|}{2n \sum_{i=1}^n \|\mathbf{x}, X_i\|_1}$$

2. **Gini Index of Projects-**

$$\text{Gini Index of Agents} = \frac{\sum_{i=1}^m \sum_{j=1}^n \|a_i - a_j\|_1}{2m \cdot \sum_{i=1}^m a_i}$$

Voting Rule	Reinforcement	Pareto-Efficiency	Monotonicity	Participation	Proportionality
R2 Mean Rule	✓ (theorem 1)	✓ (theorem 4)	✓ (theorem 7)	✓ (theorem 12)	✓ (theorem 15)
Normalized Median Rule	✓ (theorem 2)	✓ (theorem 5)	✓ (theorem 8)	✓ (theorem 13)	× (example 6)
Midpoint Rule	✓ (theorem 3)	✓ (theorem 6)	✓ (theorem 9)	✓ (theorem 14)	× (example 7)
Independent Market	✓ [?] [Theorem 5.9]	× for $m \geq n^2$ [?] [Theorem 6.1]	✓ [?] [Theorem 4.6]	✓ [?] [Theorem 5.8]	✓ [?] [Theorem 5.7]
Majoritarian Phantom	✓ [?] [Theorem 6.6]	✓ [?] [Theorem 6.1]	✓ [?] [Theorem 4.6]	✓ [?] [Theorem 6.5]	× [?] [Theorem 5.7]
R1 Quadratic	✓ (corollary 1)	× (example 2)	✓ (corollary 4)	× (example 4)	× (example 8)
R3 Quorum Median	× (example 1)	× (example 3)	✓ (theorem 10)	× (example 5)	× (example 9)
R4a Simplified Capped Median	× (corollary 2)	× (corollary 3)	✓ (theorem 11)	× (corollary 5)	× (corollary 6)

Table 1: Properties of Voting Rules (Part 1)

Voting Rule	Social Welfare Maximization	Strategyproofness	Group Strategyproofness
R2 Mean Rule	× (theorem 25)	× (theorem 16)	× (corollary 8)
Normalized Median Rule	× (theorem 26)	× (theorem 17)	× (corollary 9)
Midpoint Rule	× (theorem 27)	× (theorem 18)	× (corollary 10)
Independent Market	× [?] [Theorem 6.1]	✓ [?] [Theorem 4.8] for $m > 2$	× (example 15)
Majoritarian Phantom	✓ [?] [Theorem 6.1]	✓ [?] [Theorem 4.8] for $m > 2$	× (example 16)
R1 Quadratic	× (theorem 30)	× (theorem 21)	× (corollary 11)
R3 Quorum Median	× (corollary 14)	× (theorem 22)	× (corollary 12)
R4a Simplified Capped Median	× (corollary 15)	× (corollary 7)	× (corollary 13)

Table 2: Properties of Voting Rules (Part 2)

3. **Egalitarian Social Welfare**- measures the ℓ_1 distance between the least satisfied voter and the funding allocation:

$$\arg \max_{i \in N} (\|\mathbf{x} - X_i\|_1)$$

- **Alignment with the ground truth**- We are interested in understanding how close is the output of a voting rule to the ground truth allocation, assuming there is one. We define a naive metric to evaluate the performance of a voting rule based on the Hamming distance between the "true values" of projects and the outcome produced by the voting rule.

Formal Definition:

- Let $\mathbf{x}^* = (a_1^*, a_2^*, \dots, a_m^*)$ be the vector representing the "true values" for m projects, where $\sum_{i=1}^m a_i^* = B$ and $\mathbf{x}^*$ is normalized to 1 (assuming we know them or that we can estimate them).
- Let each agent's preference vector have an epsilon (ϵ) Hamming distance from the vector T .
- Run the voting rule to obtain the outcome vector $\mathbf{x} = (a_1, a_2, \dots, a_m)$.
- Quantify the Hamming distance between $\mathbf{x}^*$ and $\mathbf{x}$ as follows:

$$\text{Hamming}(\mathbf{x}^*, \mathbf{x}) = \sum_{i=1}^m \delta(a_i^* \neq a_i)$$

where $\delta(t_i \neq s_i)$ is an indicator function that equals 1 if $t_i \neq s_i$ and 0 otherwise.

The lower the Hamming distance, the better the voting rule performs in terms of aligning with the "true values" of the projects.

7 Theoretical Results

We provide some theoretical results.

7.1 Reinforcement

Theorem 1. *The R2 Mean Rule satisfies the reinforcement property.*

Proof. Consider two voting profiles V_1 and V_2 , each with n_1 and n_2 voters, respectively. The allocation to project p under V_1 is:

$$a_p^1 = \frac{\sum_{i \in N_1} x_{i,p}}{\sum_{i \in N_1} \sum_{p' \in P} x_{i,p'}}$$

and under V_2 is:

$$a_p^2 = \frac{\sum_{i \in N_2} x_{i,p}}{\sum_{i \in N_2} \sum_{p' \in P} x_{i,p'}}$$

When combining the two profiles into $V_3 = V_1 \cup V_2$, the allocation is:

$$a_p^3 = \frac{\sum_{i \in N_1 \cup N_2} x_{i,p}}{\sum_{i \in N_1 \cup N_2} \sum_{p' \in P} x_{i,p'}}$$

Since the sum of allocations is linear, the relative proportions between projects p and q are preserved, assuming the individual allocations in V_1 and V_2 are consistent in ranking p over q . Therefore, $a_p^3 \geq a_q^3$ if $a_p^1 \geq a_q^1$ and $a_p^2 \geq a_q^2$.

Thus, the R2 Mean Rule satisfies the reinforcement property. $\square$

Theorem 2. *The Normalized Median Rule satisfies the reinforcement property.*

Proof. Consider two voting profiles V_1 and V_2 , each with n_1 and n_2 voters respectively. Let a_p^1 be the median allocation to project p under V_1 , and a_p^2 the median allocation under V_2 . The combined profile $V_3 = V_1 \cup V_2$ has $n_1 + n_2$ voters.

Given that $a_p^1 \geq a_q^1$ and $a_p^2 \geq a_q^2$, the median allocation in the combined profile V_3 will still satisfy $a_p^3 \geq a_q^3$, because the median, as a central tendency, will reflect the relative positions from both V_1 and V_2 .

Thus, the Normalized Median Rule satisfies the reinforcement property, as combining profiles does not change the relative ordering of projects. $\square$

Theorem 3. *The Midpoint Rule satisfies the reinforcement property.*

Proof. Consider the voter v that minimizes the ℓ_1 distance in V_1 and in V_2 . Thus, $v = \operatorname{argmin} \sum_{V_1} \ell_1$ and also $v = \operatorname{argmin} \sum_{V_2} \ell_1$; therefore, $v = \operatorname{argmin} \sum_{V_1 \cup V_2} \ell_1$.

Thus, the Midpoint Rule satisfies the reinforcement property. $\square$

Corollary 1. *The Quadratic Rule satisfies the reinforcement property.*

Proof. Follows from Theorem 1 as it is a mean after taking the square root. $\square$

Example 1. *The R3 Quorum Median Rule does not satisfy the reinforcement property.*

Consider two projects, p_1 and p_2 , and let V_1 contain only v_1 , voting $[0.9, 0.1]$, and let V_2 contain only v_2 , voting $[0.9, 0.1]$. Let the quorum be 0.2. Now, the winner of V_1 is $[1, 0]$ and the winner of V_2 is $[1, 0]$, but the winner of $V_1 \cup V_2$ is $[0.8, 0.2]$.

Corollary 2. *The R4a Simplified Capped Median rule does not satisfy reinforcement since it includes a Quorum K3.*

7.2 Pareto Efficiency

Theorem 4. *The R2 Mean Rule satisfies Pareto efficiency.*

Proof. The R2 Mean Rule distributes the budget B proportionally to the sum of individual allocations.

Suppose an alternative allocation a'_p makes one voter better off by increasing the allocation for project p . Given the budget constraint B , increasing a'_p would require decreasing the allocation to another project q , thus making voters who prefer q worse off.

Since any deviation from the mean allocation would reduce the utility of at least one voter, no Pareto improvement is possible.

Therefore, the R2 Mean Rule satisfies Pareto efficiency. $\square$

Theorem 5. *The Normalized Median Rule satisfies Pareto efficiency.*

Proof. Consider a voting profile $V = \{x_{1,p}, x_{2,p}, \dots, x_{n,p}\}$ for a project p . The median allocation a_p is the median value among all voters' allocations for that project.

To prove Pareto efficiency, suppose there exists an alternative allocation a' that makes at least one voter better off without making any other voter worse off.

If we try to increase the allocation for one project p to make a particular voter better off, we must decrease the allocation for another project q due to the budget constraint.

Since the median allocation minimizes the sum of absolute deviations from voters' allocations, any reallocation that deviates from this median would necessarily make some voters worse off.

Hence, no alternative allocation can make one voter better off without making another voter worse off. Therefore, the Median Rule satisfies Pareto efficiency. $\square$

Theorem 6. *The Midpoint Rule satisfies Pareto efficiency.*

Proof. Consider the output p ; as p equals the vote of some voter v , it follows that any other p' will not equal v , thus v will be less satisfied.

Thus, the Midpoint Rule satisfies Pareto efficiency. $\square$

Example 2. *The Quadratic Rule does not satisfy Pareto efficiency.*

Consider two projects and one voter voting $[0.7, 0.3]$. As the result is the mean of the square roots, it is not equal to $[0.7, 0.3]$, which is better for the voter.

Example 3. *The R3 Quorum Median Rule does not satisfy Pareto efficiency.*

Consider two projects and one voter voting $[0.7, 0.3]$ and let the quorum be 0.4. The output will be $[1, 0]$, which is worse for the voter than $[0.7, 0.3]$.

Corollary 3. *The R4a Simplified Capped Median rule does not satisfy Pareto efficiency since it includes a Quorum K3.*

7.3 Monotonicity

Theorem 7. *The R2 Mean Rule satisfies monotonicity.*

Proof. Consider a voter i who increases their allocation to project p from $x_{i,p}$ to $x'_{i,p} > x_{i,p}$. The mean allocation to project p is given by:

$$a_p = \frac{\sum_{i \in N} x_{i,p}}{\sum_{i \in N} \sum_{p' \in P} x_{i,p'}}$$

Increasing $x_{i,p}$ increases the numerator, leading to an increase in a_p , while the denominator also increases, but not enough to reduce a_p .

Thus, increasing $x_{i,p}$ cannot decrease the final allocation a_p .

Therefore, the R2 Mean Rule satisfies monotonicity. $\square$

Theorem 8. *The Normalized Median Rule satisfies monotonicity.*

Proof. Consider a voting profile $V = \{x_{1,p}, \dots, x_{n,p}\}$ for a project p , and suppose voter i increases their allocation from $x_{i,p}$ to $x'_{i,p} > x_{i,p}$.

If $x_{i,p}$ is below the median, increasing it can shift the median upward, increasing a_p . If $x_{i,p}$ is above the median, increasing it will not decrease the median and hence will not decrease the allocation a_p . If $x_{i,p}$ is exactly at the median, increasing it will either leave the median unchanged or shift it upwards.

In all cases, increasing a voter's allocation to a project does not decrease the final median allocation. Therefore, the Normalized Median Rule satisfies monotonicity. $\square$

Theorem 9. *The Midpoint Rule satisfies monotonicity.*

Proof. Consider v , the winner. Consider a bribery that moves to some project p and consider some voter v' that gives less to p and, towards contradiction, will be the next winner. However, the distance it has is at least increasing for p (because ℓ_1 is linear); thus, contradiction.

Therefore, the Midpoint Rule satisfies monotonicity. $\square$

Corollary 4. *The Quadratic Rule satisfies monotonicity following Theorem 7.*

Theorem 10. *The R3 Quorum Median Rule satisfies monotonicity.*

Proof. There are several cases to consider:

1. If the current project is below quorum, it can only move above it.
2. If the project being reduced was above quorum but now falls below it, the project receiving additional votes can only increase since there are fewer projects above quorum to split the votes.

Therefore, the R3 Quorum Median Rule satisfies monotonicity. $\square$

Theorem 11. *The R4a Simplified Capped Median rule satisfies monotonicity.*

Proof. Considering capping with $K2$ If the current allocation of a project is below $K2$, then adding more tokens from a voter can only increase its allocation, benefiting the project. If the project's allocation is already at or above $K2$, adding more tokens will not change the allocation, as it is capped at $K2$. Therefore, the project's allocation neither decreases nor increases beyond $K2$.

Considering quorum $K3$:

1. If the current allocation of a project is below the quorum $K3$, adding more tokens can only raise the project's allocation above the quorum.
2. If a project that was above quorum $K3$ receives fewer tokens and falls below quorum, the project receiving additional votes will see its allocation increase. This occurs because there are now fewer projects above the quorum, allowing for a higher distribution of tokens to those still receiving votes.

Thus, the R4a Simplified Capped Median rule ensures that adding tokens to a project either increases or maintains its allocation, satisfying the monotonicity property. $\square$

7.4 Participation

Theorem 12. *The R2 Mean Rule satisfies the participation property.*

Proof. Consider a voter i who abstains from voting. The allocation to each project p is recalculated as:

$$a'_p = \frac{\sum_{j \in N \setminus \{i\}} x_{j,p}}{\sum_{j \in N \setminus \{i\}} \sum_{p' \in P} x_{j,p'}}$$

If voter i abstains, their influence on the allocation is removed, which generally results in a less favorable outcome for them, as their preferred projects might receive less allocation.

Thus, abstaining does not lead to a higher utility.

Therefore, the R2 Mean Rule satisfies the participation property. $\square$

Theorem 13. *The Normalized Median Rule the participation property.*

Proof. Consider a voting profile $V = \{x_{1,p}, \dots, x_{n,p}\}$ and let voter i consider abstaining. The current median allocation to project p is $a_p = \text{median}(x_{1,p}, \dots, x_{n,p})$.

If $x_{i,p}$ is below the median, abstaining does not change the median or could increase it, leading to a worse or neutral outcome for the voter. If $x_{i,p}$ is above the median, abstaining could lower the median, reducing the voter's preferred allocation. If $x_{i,p}$ is exactly at the median, abstaining would shift the median, potentially lowering the allocation for the voter's preferred project.

In all scenarios, abstaining from voting does not result in a better outcome for the voter. Therefore, the Normalized Median Rule satisfies the participation property. $\square$

Theorem 14. *The Midpoint Rule satisfies the participation property.*

Proof. There are two cases: if, by participating, the voter becomes the winner, then the voter will obviously benefit; otherwise, it is similar to the proof for the R2 Mean Rule, where the voter's participation helps minimize the sum of ℓ_1 distances.

Therefore, the Midpoint Rule satisfies the participation property. $\square$

Example 4. *The Quadratic Rule does not satisfy the participation property.*

Consider an election without a voter v where the output without v is $[0.8, 0.2]$. If the true vote of v is $[0.8, 0.2]$, then by voting, v actually pushes the mean allocation away from the preferred allocation due to the square roots.

Example 5. *The R3 Quorum Median Rule does not satisfy the participation property.*

Consider an election without a voter v where, without v , some project p_2 is not funded (but is very close to being funded). Now, if voter v votes $[1 - \epsilon, \epsilon]$ so that project p_2 receives funding, this actually gives a lot of funding to p_2 , pushing the winning allocation away from what v prefers.

Corollary 5. *The R4a Simplified Capped Median rule does not satisfy participation since it includes a Quorum $K3$.*

7.5 Proportionality

Theorem 15. *The R2 Mean Rule satisfies proportionality.*

Proof. The R2 Mean Rule aggregates votes by summing the values across all voters. This method ensures that each project receives funding proportional to the total amount of tokens allocated to it. Since the allocation directly reflects the proportion of votes each project receives, the rule satisfies proportionality. $\square$

Example 6. *The Normalized Median Rule does not satisfy proportionality.*

Consider a scenario with three voters whose vote distributions are $[1, 0]$, $[1, 0]$, and $[0, 1]$. The median allocation would be $[1, 0]$. However, proportionality requires that the allocation reflects the distribution of preferences among voters, which would result in an allocation closer to $[2/3, 1/3]$. Thus, the Normalized Median Rule fails to satisfy proportionality.

Example 7. *The Midpoint Rule does not satisfy proportionality.*

Using the same example with three voters voting $[1, 0]$, $[1, 0]$, and $[0, 1]$, the Midpoint Rule selects the input allocation that minimizes the ℓ_1 distance, hence $[1, 0]$ instead of the desired allocation of $[2/3, 1/3]$. Thus, the Midpoint Rule does not satisfy proportionality.

Example 8. *The Quadratic Rule does not satisfy proportionality.*

Consider a scenario with three voters whose vote distributions are $[9, 1]$, $[5, 5]$, and $[4, 6]$. The rule would output $[5.6, 4.4]$, where the proportional output should be $[6, 4]$.

Example 9. *The R3 Quorum Median Rule does not satisfy proportionality.*

Using the same example with three voters voting $[1, 0]$, $[1, 0]$, and $[0, 1]$, and a quorum of a minimum of 2 voters and 1.1 tokens, we get an output of $[1, 0]$ that does not satisfy proportionality.

Corollary 6. *The R4a Simplified Capped Median rule does not satisfy proportionality since it includes a Quorum K3 and is median based.*

7.6 Strategyproofness

Theorem 16. *The R2 Mean Rule does not satisfy strategyproofness.*

Example 10. Consider a scenario with two voters whose vote distributions are $[0.75, 0.25]$ and $[0, 1]$. The R2 Mean Rule will output $[0.375, 0.625]$, meaning the utility of voter 1 (measured using the ℓ_1 distance between the output and the vote) would be 0.75. If voter 1 votes untruthfully with $[1, 0]$, the R2 Mean Rule would output $[0.5, 0.5]$, reducing voter 1's ℓ_1 distance to 0.5.

Theorem 17. *The Normalized Median Rule does not satisfy strategyproofness.*

Example 11. Consider a scenario with three voters whose vote distributions are $[0.57, 0.24, 0.19]$, $[0.39, 0.48, 0.13]$, and $[0.44, 0.09, 0.48]$. The Normalized Median Rule will output $[0.506, 0.276, 0.218]$, meaning the utility of voter 1 (measured using the ℓ_1 distance between the output and the vote) would be 0.1285. If voter 1 votes untruthfully with $[0.6, 0.2, 0.2]$, the Normalized Median Rule would output $[0.524, 0.238, 0.238]$, reducing voter 1's ℓ_1 distance to 0.0962.

Theorem 18. *The Midpoint Rule does not satisfy strategyproofness.*

Example 12. Consider a scenario with three voters whose vote distributions are $[0.9, 0.1]$, $[0.4, 0.6]$, and $[0.2, 0.8]$. The Midpoint Rule will output $[0.4, 0.6]$, meaning the utility of voter 1 (measured using the ℓ_1 distance between the output and the vote) would be 1. If voter 1 votes untruthfully with $[0.5, 0.5]$, the Midpoint Rule would output $[0.5, 0.5]$, reducing voter 1's ℓ_1 distance to 0.8.

Theorem 19. *The Independent Markets Rule satisfies strategyproofness.*

Proof. The proof follows from [?][Theorem 4.8] for $m > 2$. $\square$

Theorem 20. *The Majoritarian Phantom Rule satisfies strategyproofness.*

Proof. The proof follows from [?][Theorem 4.8] for $m > 2$. $\square$

Theorem 21. *The Quadratic Rule does not satisfy strategyproofness.*

Example 13. Consider a scenario with three voters whose vote distributions are $[0.7, 0.3]$, $[0.4, 0.6]$, and $[0.3, 0.7]$. The Quadratic Rule will output $[0.483, 0.517]$, meaning the utility of voter 1 (measured using the ℓ_1 distance between the output and the vote) would be 0.434. If voter 1 votes untruthfully with $[0.8, 0.2]$, the Quadratic Rule would output $[0.502, 0.498]$, reducing voter 1's ℓ_1 distance to 0.396.

Theorem 22. The R3 Quorum Median Rule does not satisfy strategyproofness.

Example 14. Consider a scenario with three voters whose vote distributions are $[0.2, 0.8]$, $[0.1, 0.9]$, and $[0.5, 0.5]$ with a quorum of 0.3 tokens and 2 voters for a project. The R3 Quorum Median Rule will output $[0, 1]$, meaning the utility of voter 1 (measured using the ℓ_1 distance between the output and the vote) would be 0.4. If voter 1 votes untruthfully with $[0, 1]$, the Quadratic Rule that filters out all 0 votes would output $[0.25, 0.75]$, reducing voter 1's ℓ_1 distance to 0.1.

Corollary 7. The R4a Simplified Capped Median rule does not satisfy Strategy proofness since it includes a Quorum K3 and is median based.

7.7 Group Strategyproofness

Corollary 8. The R2 Mean Rule does not satisfy group strategyproofness, as it does not satisfy strategyproofness.

Corollary 9. The Normalized Median Rule does not satisfy group strategyproofness, as it does not satisfy strategyproofness.

Corollary 10. The Midpoint Rule does not satisfy group strategyproofness, as it does not satisfy strategyproofness.

Theorem 23. The Independent Markets Rule does not satisfy group strategyproofness.

Example 15. Consider a scenario where a group of three voters cast their votes as $[0.01, 0.99]$, $[0.97, 0.03]$, and $[0.46, 0.54]$.

The Independent Markets algorithm computes an allocation of approximately $[0.4781, 0.5219]$. The ℓ_1 distances (which represent the dissatisfaction of the voters with this outcome) between each voter's preference and the computed allocation are 0.9398, 0.9803, and 0.0439, respectively.

However, if the first and third voters decide to misreport their preferences as $[0.4, 0.6]$ and $[0.4, 0.6]$, while the second voter remains truthful, the algorithm would compute a new allocation of $[0.45, 0.55]$. This new allocation decreases their respective ℓ_1 distances to 0.8836, 1.0364, and 0.0123.

Intuitive Explanation: - The Independent Markets Rule tries to balance the preferences of all voters by considering "phantom" voters. - In this example, the true preferences are very skewed: one voter heavily favors the first project, another favors the second, and the third voter is more balanced. - Initially, the rule finds a compromise between these preferences, but it isn't perfect for anyone. - When the first and third voters misreport their preferences to show more moderate support, the algorithm shifts the allocation slightly closer to their new reported preferences. This reduces the overall dissatisfaction (as measured by ℓ_1 distance) for the first and third voters, even though the second voter becomes slightly more dissatisfied. - This shows that groups of voters can strategically manipulate their reports to achieve a more favorable outcome, demonstrating that the Independent Markets Rule is not group strategyproof.

Theorem 24. The Majoritarian Phantom Rule does not satisfy group strategyproofness.

Example 16. Consider a scenario where three voters cast their votes as $[0.93, 0.07]$, $[0.73, 0.27]$, and $[0.99, 0.01]$.

The Majoritarian Phantom algorithm computes an allocation of approximately $[0.8303, 0.1697]$. The ℓ_1 distances between each voter's preference and the computed allocation are 0.2049, 0.2049, and 0.3287, respectively.

However, if the first and third voters decide to misreport their preferences as $[1, 0]$ and $[1, 0]$, while the second voter remains truthful, the algorithm would compute a new allocation of $[0.8639, 0.1361]$. This new allocation decreases their respective ℓ_1 distances to 0.1377, 0.2722, and 0.2614.

Intuitive Explanation: - The Majoritarian Phantom Rule tries to reflect the majority preferences while also considering the intensity of these preferences. - In this example, the original preferences are strongly in favor of the first project, but there is some support for the second project. - Initially, the rule gives a significant allocation to the first project but still provides some allocation to the second. - When the first and third voters misreport their preferences to show complete support for the first project, the

algorithm shifts the allocation even more toward the first project. - This manipulation results in a slightly better outcome for the first and third voters, even though the second voter, who is more balanced, becomes more dissatisfied. - This example shows that groups of voters can influence the outcome by strategically misreporting their preferences, proving that the Majoritarian Phantom Rule is not group strategyproof.

Corollary 11. *The Quadratic Rule does not satisfy group strategyproofness, as it does not satisfy strategyproofness.*

Corollary 12. *The R3 Quorum Median Rule does not satisfy group strategyproofness, as it does not satisfy strategyproofness.*

Corollary 13. *The R4a Simplified Capped Median rule does not satisfy Strategy proofness since it does not satisfy strategyproofness.*

7.8 Maximal Social Welfare (Minimal L1)

A solution to an instance satisfies maximum SWF if there is no other allocation that results in a smaller total ℓ_1 distance between the final allocation and the ballots.

Theorem 25. *The R2 Mean Rule does not satisfy maximum SWF.*

Example 17. *Consider the following example:*

Setup:

- **Projects:** P_1, P_2
- **Voters:** 3 voters (A, B, C)
- **Votes:**
 - **Voter A:** (1, 0)
 - **Voter B:** (1, 0)
 - **Voter C:** (0.2, 0.8)

In this setup, Voters A and B strongly favor P_1 , while Voter C strongly favors P_2 .

R2 Mean Rule Calculation:

- **Mean Allocation:**

$$a = \left(\frac{1 + 1 + 0.2}{3}, \frac{0 + 0 + 0.8}{3} \right) = \left(\frac{2.2}{3}, \frac{0.8}{3} \right) \approx (0.7333, 0.2667)$$

L_1 Distance Calculation for R2 Mean Rule:

- **Voter A:** $|1 - 0.7333| + |0 - 0.2667| = 0.2667 + 0.2667 = 0.5334$
- **Voter B:** $|1 - 0.7333| + |0 - 0.2667| = 0.2667 + 0.2667 = 0.5334$
- **Voter C:** $|0.2 - 0.7333| + |0.8 - 0.2667| = 0.5333 + 0.5333 = 1.0666$
- **Total L_1 Distance:** $0.5334 + 0.5334 + 1.0666 = 2.1334$

Midpoint Rule as an Alternative:

- **Midpoint Allocation:** (1, 0)

L_1 Distance Calculation for Midpoint Allocation:

- **Voter A:** $|1 - 1| + |0 - 0| = 0 + 0 = 0$
- **Voter B:** $|1 - 1| + |0 - 0| = 0 + 0 = 0$
- **Voter C:** $|0.2 - 1| + |0.8 - 0| = 0.8 + 0.8 = 1.6$
- **Total L_1 Distance:** $0 + 0 + 1.6 = 1.6$

Remark 2. The R2 Mean Rule satisfies maximum SWF if it is defined as minimizing the ℓ_2 distance between the aggregation output and all ballots [?].

Theorem 26. The Normalized Median Rule does not satisfy maximum SWF optimization.

Example 18. Consider the following example with two projects and three voters:

Votes:

- **Voter A:** (0.1, 0.9)
- **Voter B:** (0.4, 0.2)
- **Voter C:** (0.6, 0.1)

Normalized Median Rule Calculation:

- Normalized Median for P_1 : 0.4
- Normalized Median for P_2 : 0.2

The sum of the medians is $0.4 + 0.2 = 0.6$. To ensure the allocations sum to 1, normalization is required:

$$(0.4, 0.2) \text{ is normalized to } \left(\frac{0.4}{0.6}, \frac{0.2}{0.6} \right) = (0.667, 0.333)$$

L_1 Distance for Normalized Median Rule:

$$\begin{aligned} \text{Voter A: } & |0.1 - 0.667| + |0.9 - 0.333| = 0.567 + 0.567 = 1.134 \\ \text{Voter B: } & |0.4 - 0.667| + |0.2 - 0.333| = 0.267 + 0.133 = 0.4 \\ \text{Voter C: } & |0.6 - 0.667| + |0.1 - 0.333| = 0.067 + 0.233 = 0.3 \\ \text{Total } L_1 \text{ Distance: } & 1.134 + 0.4 + 0.3 = 1.834 \end{aligned}$$

Alternative Allocation: Consider the allocation (0.5, 0.5).

L_1 Distance for Alternative Allocation:

$$\begin{aligned} \text{Voter A: } & |0.1 - 0.5| + |0.9 - 0.5| = 0.4 + 0.4 = 0.8 \\ \text{Voter B: } & |0.4 - 0.5| + |0.2 - 0.5| = 0.1 + 0.3 = 0.4 \\ \text{Voter C: } & |0.6 - 0.5| + |0.1 - 0.5| = 0.1 + 0.4 = 0.5 \\ \text{Total } L_1 \text{ Distance: } & 0.8 + 0.4 + 0.5 = 1.7 \end{aligned}$$

Remark 3. The Median Rule satisfies maximum SWF if it is defined as minimizing the ℓ_1 distance between the aggregation output and all ballots and when there is no normalization of the final allocation [?].

Theorem 27. The Midpoint Rule does not satisfy maximum SWF.

Example 19. Consider 4 projects (P_1, P_2, P_3, P_4) and 4 voters with the following votes:

$$\begin{aligned} \text{Voter A: } & (0.8, 0.1, 0.05, 0.05) \\ \text{Voter B: } & (0.1, 0.8, 0.05, 0.05) \\ \text{Voter C: } & (0.05, 0.05, 0.8, 0.1) \\ \text{Voter D: } & (0.05, 0.05, 0.1, 0.8) \end{aligned}$$

The Midpoint Rule might select any voter allocation since all ballots have the same distance from one another. The total ℓ_1 distance would always be 4.6.

Consider the alternative allocation (0.25, 0.25, 0.25, 0.25). The total ℓ_1 distance will now be 4.4.

Example 20. Consider the following example: we have 3 voters, each giving a ballot for two projects: (0.99, 0.01), (0.01, 0.99), (0.2, 0.8).

Theorem 28. The IM algorithm does not satisfy maximum SWF as following the proof of [?]/[Theorem 6.1].

Theorem 29. The Majoritarian Phantoms algorithm satisfies maximum SWF, following the proof of [?]/[Theorem 6.1].

Theorem 30. *The Quadratic Voting rule does not satisfy Social Welfare.*

Example 21. Setup:

- **Projects:** P_1, P_2
- **Voters:** 3 voters (A, B, C)
- **Votes:**
 - **Voter A:** (0.01, 0.99)
 - **Voter B:** (0.01, 0.99)
 - **Voter C:** (0.99, 0.01)

In this example, two voters strongly favor P_2 , while one voter strongly favors P_1 .

Quadratic Voting Calculation:

- **True Votes:**
 - **Voter A:** $\sqrt{0.01} \approx 0.1, \sqrt{0.99} \approx 0.995$
 - **Voter B:** $\sqrt{0.01} \approx 0.1, \sqrt{0.99} \approx 0.995$
 - **Voter C:** $\sqrt{0.99} \approx 0.995, \sqrt{0.01} \approx 0.1$
- **Aggregate Votes:**
 - For P_1 : $0.1 + 0.1 + 0.995 = 1.195$
 - For P_2 : $0.995 + 0.995 + 0.1 = 2.09$
- **Final Allocation:**

$$a_1 = \frac{1.195}{3.285} \approx 0.364, \quad a_2 = \frac{2.09}{3.285} \approx 0.636$$

L_1 Distance Calculation for Quadratic Voting:

- **Voter A:** $|0.01 - 0.364| + |0.99 - 0.636| = 0.354 + 0.354 = 0.708$
- **Voter B:** $|0.01 - 0.364| + |0.99 - 0.636| = 0.354 + 0.354 = 0.708$
- **Voter C:** $|0.99 - 0.364| + |0.01 - 0.636| = 0.626 + 0.626 = 1.252$
- **Total L_1 Distance:** $0.708 + 0.708 + 1.252 = 2.668$

Alternative Allocation:

Consider an alternative allocation that better aligns with the majority preference: (0.2, 0.8).

L_1 Distance Calculation for Alternative Allocation:

- **Voter A:** $|0.01 - 0.2| + |0.99 - 0.8| = 0.19 + 0.19 = 0.38$
- **Voter B:** $|0.01 - 0.2| + |0.99 - 0.8| = 0.19 + 0.19 = 0.38$
- **Voter C:** $|0.99 - 0.2| + |0.01 - 0.8| = 0.79 + 0.79 = 1.58$
- **Total L_1 Distance:** $0.38 + 0.38 + 1.58 = 2.34$

Corollary 14. *The R3 Quorum Median Rule does not satisfy maximum SWF since the Median does not satisfy maximum SWF (because of the normalization).*

Corollary 15. *The R4a Simplified Capped Median rule does not satisfy maximum SWF since the Median does not satisfy maximum SWF (because of the normalization).*

8 Experimental Design (Work in Progress!)

First we discuss the artificial cumulative vote generation model we are using, that has been shown in previous works to be similar to real life scenarios [?].

8.1 Mallows Model For Vote Generation

The traditional Mallows model generates rankings based on a central reference ranking, often called the base vote, with a dispersion parameter controlling variability. Our approach is similar, but instead of generating rankings, we distribute tokens across projects.

In this model, we generate a matrix of cumulative votes for n voters and m projects. Each voter's total vote sums to K (in our case $K = 1$), and the votes are generated with some randomness controlled by the parameter α .

We begin by generating a *base vote* for all projects using the Dirichlet distribution. This ensures that the sum of votes for each voter is exactly K . Noise is then introduced into each voter's vote by blending the base vote with an independent Dirichlet sample, with α controlling the degree of noise.

Formally Let V_i represent the vote of voter i , where $i \in \{1, 2, \dots, n\}$, and let m be the number of projects. The goal is to ensure that the sum of votes for each voter satisfies:

$$\sum_{j=1}^m V_{i,j} = 1, \quad \text{for each } i = 1, 2, \dots, n.$$

The model is based on two components:

1. A **base vote** B generated from a Dirichlet distribution:

$$Base \sim \text{Dirichlet}(\mathbf{1}_m),$$

where $\mathbf{1}_m$ is a vector of ones of size m , and the base vote is scaled so that it sums to K .

2. A **noisy vote** V_i for each voter i is generated by taking a weighted combination of the base vote B and an independent sample from another Dirichlet distribution, controlled by α :

$$V_i = (1 - \alpha) \cdot Base + \alpha \cdot \text{Dirichlet}(\mathbf{1}_m).$$

Here, $\alpha \in [0, 1]$ controls the level of noise:

- If $\alpha = 0$, each voter's vote is exactly the base vote B (homogeneous).
- If $\alpha = 1$, each voter's vote is independently generated (heterogeneous).
- For values of α between 0 and 1, the votes are a mixture of homogeneity (shared base vote) and heterogeneity (individual noise).

We use $\alpha = 0.5$ which represents a balance between **homogeneity** (all voters being similar in their vote distributions) and **heterogeneity** (each voter having individual differences).

References